```python
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

array = np.random.randint(1, 100, 16)
print(array)

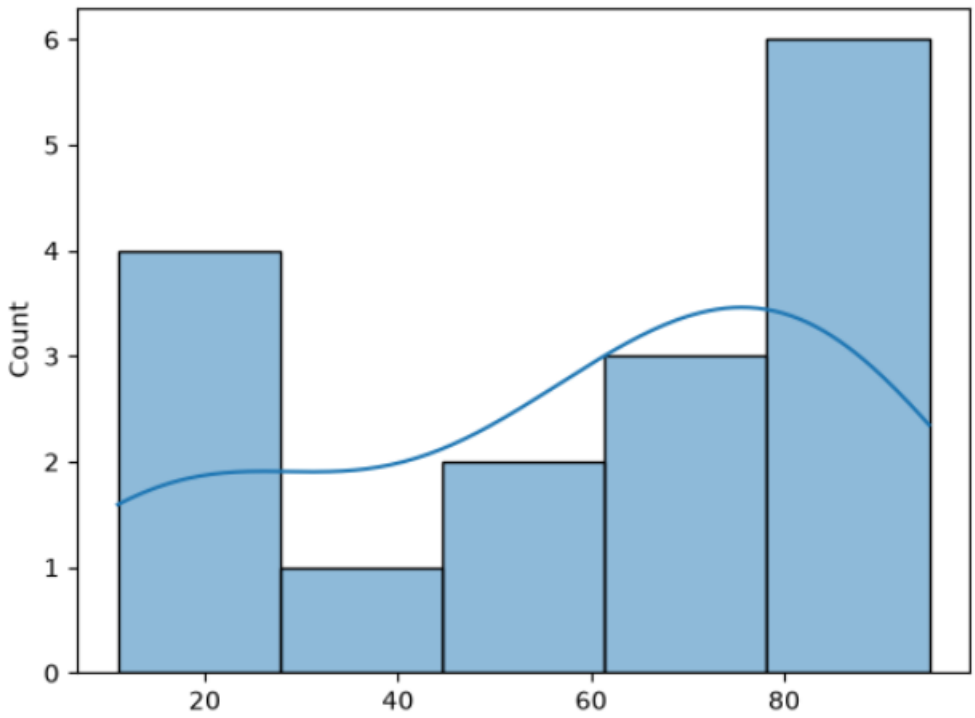
print(array.mean())
print(np.percentile(array, 25))
print(np.percentile(array, 50))
print(np.percentile(array, 75))
print(np.percentile(array, 100))
```

```
[63 42 19 84 20 51 95 58 67 87 70 80 81 87 11 19]
58.375
36.5
65.0
81.75
95.0
```

```python
def outDetection(array):
    Q1, Q3 = np.percentile(array, [25, 75])
    IQR = Q3 - Q1
    lr = Q1 - (1.5 * IQR)
    ur = Q3 + (1.5 * IQR)
    return lr, ur

lr, ur = outDetection(array)
print(lr, ur)
sns.histplot(array, kde=True)
plt.show()
```

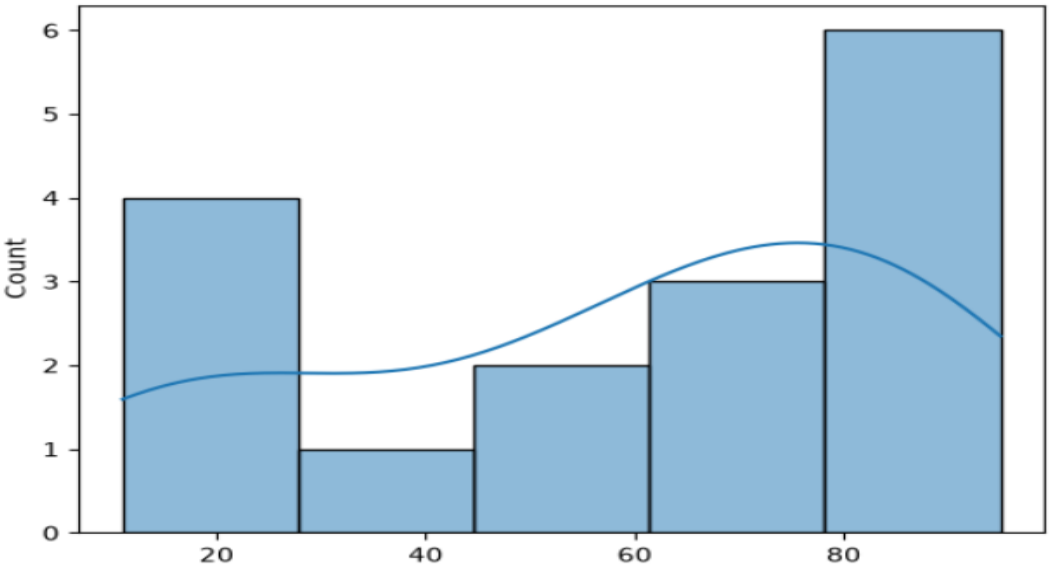
```
-31.375 149.625
```



```
new_array = array[(array > lr) & (array < ur)]
print(new_array)

sns.histplot(new_array, kde=True)
plt.show()
```

```
[63 42 19 84 20 51 95 58 67 87 70 80 81 87 11 19]
```




```
lr1, ur1 = outDetection(new_array)
print(lr1, ur1)
final_array = new_array[(new_array > lr1) & (new_array < ur1)]
print(final_array)
sns.histplot(final_array, kde=True)
plt.show()
```

```
-31.375 149.625
[63 42 19 84 20 51 95 58 67 87 70 80 81 87 11 19]
```

